The Room Is Part of the Instrument

Sixty-nine documents on anechoic chambers, from a 1945 wartime report to a 2026 drone-directivity paper, tell one story when read in sequence: the room's rating is a property of its lining, its real limit is set by whatever you put inside it, and whether you can see the difference depends on how you look.

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01 1945: a rule that was never about the room

The story starts in a Harvard laboratory during the war. Beranek, Sleeper and Moots, writing OSRD Report 4190 in 1945, tested wedge after wedge and found that "the lowest frequency at which good absorption is obtained, is that for which the length of the wedge equals approximately one-fourth of a wave length". They were explicit about what that costs: "for absorptions down as low as 40 cps, a wedge structure would need to have a total length of 94 inches measured from its tip to the rigid backing wall". And they were equally explicit about what happens when absorption is not perfect: "the reflected wave will combine with the incident wave to form a standing wave pattern". OSRD 4190, §III.C.5, pp. 28–29

Everything that follows is a footnote to those two sentences. The quarter-wave rule became the number on every chamber's brochure. The standing-wave sentence became the thing every later paper rediscovers, because the rule describes a wedge in a tube and the room is not a tube.

02 The number on the brochure and the number in the room

Seventy years later the gap between the two is measured routinely. Jiang, Zhang and Huang put the mechanism in one line: "99% energy absorption means 10% pressure reflection, which is still very large. In some area inside the anechoic chamber, the multi-reflected sound accumulates and violates the 1.5 dB deviation easily." In their 9 × 8 × 7 m simulation, wedges optimised for a 100 Hz cut-off kept only 56.7 % of the centre-to-vertex line inside the ISO tolerance at 100 Hz; the room's effective cut-off was 128 Hz. JSV 381, 2016, pp. 150–153

Real rooms behave the same way. Schneider measured the LMA Marseille chamber, 5.4 × 6.3 × 11.4 m with 0.7 m melamine wedges, on a 0.2 m grid with pure tones and found "significant deviations from free-field conditions occur in the 100–160 Hz frequency range", plus an asymmetry between 170 and 190 Hz that no uniform model could reproduce. Haasjes, reviewing that case, states the consequence: a chamber "designed with a cut off frequency of 75 Hz" has a real cut-off that "lies at 200 Hz rather than 75 Hz", with the lined doors among the causes. His general verdict covers the field: chambers "suffer from low-frequency sound waves (up to about 200 Hz) reflecting off the walls, in some cases even above the cut off frequency the chamber is designed for". JSV 320, 2009, pp. 993–994; Haasjes 2025, pp. 3–4

Small rooms show the collapse in the raw numbers. In Singh, Garg and Narayanan's 2.6 × 1.7 × 2.2 m chamber the level fell 5.4–6.2 dB per doubling of distance at 315 Hz, 3.0–3.7 dB at 200 Hz, and 0.01–0.30 dB at 100 Hz: at 100 Hz the sound was simply not getting quieter with distance. In Garg's 2 m³ calibration box the deviation at 125 Hz swung from +4.52 dB at 0.64 m to −7.13 dB at 1.14 m, leaving a usable space "between 79 and 89 cm". Even a well-designed facility slips: TU Delft's tunnel plenum was laid out for free field "above approximately 173.5 Hz" and its own conclusion reports "a cutoff frequency of 200 Hz". IJA 19, 2020, Tables 2–3; MAPAN 34, 2019, Table 2; Appl. Acoust. 170, 2020, pp. 4, 14

> **Below the cut-off a free field does survive close to the source. Bonfiglio and Pompoli, comparing image-source predictions with two large chambers, found that "the frequencies lower than the cut-off limit still guarantee a free field condition in a reduced volume of the chamber up to a distance of around 1 m from the boundaries".**
> JASA 134, 2013, p. 291

So the brochure number is a lower bound on trouble, not an upper bound. Zawodny and Haskin give the operational rule that falls out of this: their microphones sat 0.49 m from the wedge tips, and "treating this as a 1/4-wavelength yields a notional cut-on frequency of 175 Hz", below which a 5 dB directivity error appeared at one angle. Whatever the room is rated, each microphone has its own limit at c/(4 × distance to the nearest wedge). AIAA 2017, p. 16

03 How you look decides what you see

If the failure is a standing-wave pattern, then the way you sample it, excite it and fit it can each hide it. Cunefare's group at Georgia Tech showed all three in 2003. Their continuous traverses revealed a deviation that "is due to interference between the direct and reflected components of the wave field", with zero crossings every 29–41 % of a wavelength. Sampling at 0.3 m, the common practice, "would indicate no performance issues with the chamber over the entire traverse" at 4 kHz where the continuous traverse found violations within 1.5 m. Their recommended spacing is 15 % of a wavelength. Exciting with band-limited noise instead of tones produced "very little deviation from inverse square law performance over these traverses, regardless of the analysis method"; at 500 Hz the tone traverse was 37 % out of tolerance and the noise traverse 0 %. And fitting with a free acoustic-centre offset cut the 80 Hz out-of-tolerance fraction from 56 % to 13 % on identical data. JASA 113, 2003, pp. 889–892, Tables IX, XI

Every later qualification study confirms one of those three. Nash's 51 m³ commercial chamber "could satisfy the ISO tolerances when using random noise but failed to qualify when using pure tones"; a 2056 Hz tone fitted −26.4 dB per decade, "a value that is physically impossible", the signature of "an acoustical cancellation occurring between the arriving wave front and a strong reflection from a nearby surface". NPL stripped a hemi-anechoic room of its wedges in eleven steps: pure-tone deviations reached 14.3 dB while broadband never exceeded 3.7 dB, "most likely due to averaging of the interference between the limits of the frequency band being analysed". Worse, the ISO 3745:2003 regression with a free offset passed "all room states … at all frequencies except 125 Hz, even when all the absorptive wedges had been removed". Rooms passing that fit carried A-weighted sound-power errors of 6.4–7.3 dB; rooms passing the older broadband rule 0.8–1.5 dB; rooms qualifying on tones about 0.02 dB. ICA 2019, pp. 1343–1349; NPL DQL-AC 007, 2004, pp. 22–51

Winker and Stahnke explain why the standards allowed it: ISO 3745 only says the fitted offset "should" fall within 200 mm, so "a chamber can fully comply with ISO 3745 at certain frequencies while not exhibiting free-field performance at those frequencies". ISO 26101 fixed the slope and made the offset a distance to be measured, but introduced its own geometric trap, a traverse of λ/2 at the lowest frequency, which limited a 3.35 × 3.05 × 2.74 m room to 214 Hz "due to geometry constraints" and the Split chamber to 250 Hz until the source was moved to a wall. Jenny and Anderson found the same fit absorbing ultrasonic air absorption into a fake acoustic centre 13.7–20.6 cm behind a tweeter, "clearly not accurate as the acoustic center cannot be located that far away from the physical surface of the transducer". Inter-Noise 2016, pp. 3–5; Euronoise 2018, p. 2229; JASA-EL 130, 2011, EL72–EL73

> **The standards now say it themselves. ISO 26101-1: qualification "shall be made using a bandwidth that is typical of the spectral characteristics of the type of devices that will be measured". ISO 5305 for drones: the tolerance must hold "for all source positions and measurement directions specified", and it concedes that band summation is what makes interference disappear.**
> ISO 26101-1:2021 §5.1.4.1; ISO 5305:2024 §6.2, §7.2.2

04 The reflector you brought with you

When a chamber fails, it is rarely the wedges. Rizzi's NASA SALT room, 337 m³, passed ISO 3745 everywhere "except the farthest measurement locations below 250 Hz", next to the corner opposite the source. Nash's failure came from an expanded-metal floor grate. Schneider's asymmetry came from doors. The ITMO group found their worst direction 15.5 dB off and blamed "the non-uniform distribution of the absorbing coating": an uncovered floor recess 0.15 × 1.3 × 3.55 m and flat ceiling patches. Winker found that above 10 kHz "the most likely factor is the influence of the measurement system inside the free-field creating reflections that do not appear at lower frequencies". RASD 2013 §3.2; ICA 2019; JSV 320 p. 994; arXiv 2603.16556 §2, §4.4; Inter-Noise 2016 pp. 7–8

For a directivity measurement the fixture is the worst offender because its error looks exactly like a lobe. González and Kob measured the same loudspeaker in four rooms and saw "pressure level reductions of up to 18 dB" at specific frequencies, "comb filtering effect … due to the very early reflections arriving from the table and other nearby surfaces" above 1 kHz, and a separate low-frequency artefact where "the measurement axis" aligned "with a node of an excited room mode" around 90 Hz. Fasulo's rotor study on the same Tyto 1585 stand we use found that the floor "redirect[s] tonal energy into directions that should otherwise be relatively quiet, most notably at 0° and 180°", that "the support pylon lies directly on the acoustic line of sight of microphone 10, producing shielding", and that "even modest geometric asymmetries, here the support pylon, can shift directivity by up to 4 dB". DAGA 2026, pp. 1221–1223; Aerospace 12:647, 2025, pp. 10–24

The floor deserves its own paragraph. Ma's group measured the same drone in the same HKUST chamber with and without the floor wedges: two adjacent microphones near the floor differed by up to 10 dB at the blade-passage frequency in the hemi-anechoic case and by little in the full anechoic one, so that "using a hemi-anechoic chamber could considerably degrade the tonal noise assessment accuracy". Rasmussen and Winberg derive why: a microphone at height h over a plane sees a comb filter whose first dip is at c/(4h), 71 Hz for a 1.2 m pole, with dips exceeding 20 dB, and "if the source signal contains pure tone components, it will not be possible to do this correction". Foam helps only at high frequency. Alkmim covered 2.5 × 2.5 m under a drone with 7 cm of it and found the reflection "minimized during measurements but not fully suppressed, especially at low frequencies"; Jawahar's 24 mm 75PPI-T treatment removed about 13 dB, but in the 1.5–10 kHz band, and even at four rotor radii the bare plate still added 4–8 dB above 1 kHz at a 50° angle while adding nothing at 90°. Quiet Drones 2022 (Ma) pp. 8, 12; Quiet Drones 2022 (Rasmussen) pp. 3–5; JASA 152, 2022, p. 2737; Sci. Rep. 15:2170, 2025, pp. 6–9

05 The rotor brings its own room

A loudspeaker leaves the air alone. A rotor does not, and a closed room hands the wake back to it. Stephenson, Weitsman and Zawodny watched it happen in NASA's Small Hover Anechoic Chamber, a sealed room of about 32 m³: for the first seconds after the rotor reached speed the spectrum showed the blade-passage frequency alone; "at approximately 7.5 s into the run, flow recirculation in the room has established and the higher harmonics of the BPF are now observable", up by "in excess of 15 dB", the second harmonic at out-of-plane microphones by about 30 dB, while the fundamental and the in-plane microphones barely moved. Their verdict: "This higher harmonic content is not indicative of what a rotor in hover outdoor would emit, but is instead an artifact of testing within the confined space of an anechoic chamber." JASA 145, 2019, pp. 1153–1155

Weitsman's follow-up quantified the cost and a partial cure. With a clean window of only 3.9 s the random uncertainty of a 2 s spectrum was ±1.1 dB; two woven meshes placed downstream at 3.3 and 6 rotor diameters stretched the window to about 10 s and halved the uncertainty, while a mesh upstream "distorts the inflow" and made things worse. Then Whelchel tried screens in a different room with different rotors and found "no beneficial effect on recirculation and turbulence ingestion", concluding that "an effective screen configuration is rotor and anechoic chamber dependent". Nardari's simulations put the confined-room penalty at about 5 dBA overall and 5–10 dB on the third harmonic and above, and their loudspeaker calibration of the same hemi-anechoic room found it within ±1 dB above 2.2 times the blade-passage frequency "but only within ±4 dB below that, primarily due to the onset of standing waves, which are a consequence of the nearly cuboidal shape of the chamber". JASA 148, 2020, pp. 1327–1335; Whelchel 2023, pp. 82–85; AIAA 2019-2497, pp. 2–16

The effect is not universal, which is the point. Gallo's bench at VKI vents the wake through a wall opening into a second room and saw no rise over 30 s; Fasulo's rotor at CIRA sat six diameters above the floor with twelve to the nearest wall and its harmonics "remain sharp and strictly horizontal". A free-flying drone in HKUST's 250 m³ chamber showed recirculation only after about 30 s and only as broadened tones and a 1–2 dB broadband rise that "the instantaneous OASPL value cannot distinguish". Nobody has a model for when it starts; every facility finds out for itself. Acta Acust. 9:16, 2025, p. 7; Aerospace 12:647, p. 18; Acoust. Aust. 52, 2024, pp. 316–321

06 Why the model will not warn you, and what does

It is tempting to simulate the room and skip the traverse. The literature says the cheap simulation cannot see the problem and the expensive one needs data nobody has. Brinkmann's round robin found that "most present simulation algorithms generate obvious model errors once the assumptions of geometrical acoustics are no longer met", that no algorithm "was able to exactly match the comb filter structure" of even a single reflection, and that low-frequency reverberation was overestimated by 58 % at 125 Hz. Götz's comparison of wave-based and ray-based simulation against measurement gave correlations of 0.73–0.92 for the wave solver and 0.10–0.56 for ray tracing. Vorländer's error-propagation analysis concludes that predicting reverberation to within a just-noticeable difference "requires input data in a quality which is not available from reverberation room measurements", and that "it is not adequate to 'calibrate' a computer model by modification of input data". JASA 145, 2019, pp. 2746–2757; arXiv 2509.05175 Table 3; JASA 133, 2013, p. 1203

What does work is narrower. Schneider showed that a flat wall with a locally-reacting admittance from a tube "failed to predict the quality of the chambers at frequencies below ~150 Hz, regardless of the admittance applied", and that three linings with absorption coefficients of 0.997, 1.0 and 0.997 gave "significant differences in the size of the 1.5 dB region"; only a non-local admittance built from a finite-element model of the wedge array, mounting frame included, recovered the measured perturbations. Bonfiglio and Jiang get useful answers without meshing the room by summing complex image sources with a spherical-wave reflection coefficient, Jiang's version with angle-dependent wedge impedance. Beyond prediction, the literature offers three ways to push the limit down: hybrid or graded linings (Wang and Tang, Jiang), source and microphone placement (Bonfiglio's 1 m, Rizzi's corners, Zawodny's quarter-wave), and active cancellation, which Haasjes brought to 9.4–9.6 dB in a real-time 2-D rig up to about 600 Hz and Friot to 10 dB at 280 Hz in a real room, at a cost Friot estimates at roughly 100 microphones and loudspeakers for 100 Hz in a 10 × 7 × 6 m chamber. And when a room fails, one compact spherical array can say why: Sun localised the direct path and five first-order reflections to 1–6° at 3–17 dB signal-to-noise, and Mabande placed walls "precisely up to a few centimeters only". JSV 320 pp. 996–1001; Haasjes 2025 pp. 112–119; arXiv 0911.4639 §3–5; JASA 131, 2012, p. 2835; JASA 134, 2013, p. 2786

07 What this means for six microphones on an arc and a rotor on a stand

The protocol the story implies

1. **Qualify with the rig in place, with tones, along the arc.** Pure tones at the blade-passage frequency and its first harmonics, traversed along each microphone's radial, plotted against a fixed −6 dB per doubling line, with the fitted acoustic-centre offset reported and rejected if implausible. A noise-only pass is evidence for nothing here (Cunefare, NPL, Nash, Winker, ISO 5305).
2. **Compute each microphone's own floor.** c/(4 × distance to the nearest wedge tip), and treat corners below 250 Hz as off limits (Zawodny, Rizzi). Expect the room's real limit well above its rating (Jiang, Schneider, Garg, Merino-Martínez).
3. **If the floor is hard, do not report tonal directivity from it.** Foam fixes the high band only; flush mounting or a fully anechoic floor are the options that survive contact with the literature (Ma, Rasmussen, Alkmim, Jawahar).
4. **Record from spin-up and find the clean window per run.** Harmonics rising while the fundamental holds, strongest off-axis, is recirculation, not the prop. Settle the air between runs. Do not import another facility's mesh or settle time (Stephenson, Weitsman, Whelchel, Ma 2024).
5. **Reverse the rotation once per configuration.** An asymmetry that flips is the stand or the room; one that stays is the prop (Fasulo).
6. **Report the envelope, not the adjective.** Frequency × distance × direction, with the signal used. "Measured in an anechoic chamber" describes the wedges, not the measurement (ITMO, Palchikovskiy, ISO 26101-1).

How this was built: 69 documents were downloaded, identified from their own front matter (papers/BIBLIOGRAPHY.md), read, extracted with page-referenced quotes, and then audited claim by claim by a second pass that found and corrected 27 errors in earlier drafts (papers/ERRATA.md). Two sources are held only as abstracts (Hanson 2023, Stroud 2010) and are not cited here. The Beranek document held is the 1945 OSRD report, not the 1946 journal article. Companion documents: the field guide to symptoms (anechoic-chamber-problems-brief.pdf), the graded evidence matrix (chamber-literature-synthesis.pdf), the simulation review (anechoic-simulation.html) and the reflection-localisation review (reflection-localization.html). SoundVisualizer, 2026-09-07.